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ENSEMBLE ADVERSARIAL TRAINING: ATTACKS AND DEFENSES

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ABSTRACT

Adversarial examples are perturbed inputs designed to fool machine learning models. Adversarial training injects such examples into training data to increase robustness. To scale this technique to large datasets, perturbations are crafted using fast *single-step* methods that maximize a linear approximation of the model's loss.

We show that this form of adversarial training converges to a degenerate global minimum, wherein small curvature artifacts near the data points obfuscate a linear approximation of the loss. The model thus learns to generate weak perturbations, rather than defend against strong ones. As a result, we find that adversarial training remains vulnerable to black-box attacks, where we *transfer* perturbations computed on undefended models, as well as to a powerful novel single-step attack that escapes the non-smooth vicinity of the input data via a small random step.

We further introduce *Ensemble Adversarial Training*, a technique that augments training data with perturbations transferred from other models. On ImageNet, Ensemble Adversarial Training yields models with stronger robustness to black-box attacks. In particular, our most robust model won the first round of the *NIPS 2017 competition on Defenses against Adversarial Attacks* (Kurakin et al., 2017c). However, subsequent work found that more elaborate black-box attacks could significantly enhance transferability and reduce the accuracy of our models.

1 INTRODUCTION

Machine learning (ML) models are often vulnerable to *adversarial examples*, maliciously perturbed inputs designed to mislead a model at test time (Biggio et al., 2013; Szegedy et al., 2013; Goodfellow et al., 2014b; Papernot et al., 2016a). Furthermore, Szegedy et al. (2013) showed that these inputs *transfer* across models: the same adversarial example is often misclassified by different models, thus enabling simple *black-box* attacks on deployed models (Papernot et al., 2017; Liu et al., 2017).

Adversarial training (Szegedy et al., 2013) increases robustness by augmenting training data with adversarial examples. Madry et al. (2017) showed that adversarially trained models can be made robust to *white-box* attacks (i.e., with knowledge of the model parameters) if the perturbations computed during training closely maximize the model's loss. However, prior attempts at scaling this approach to ImageNet-scale tasks (Deng et al., 2009) have proven unsuccessful (Kurakin et al., 2017b).

It is thus natural to ask whether it is possible, at scale, to achieve robustness against the class of *black-box* adversaries Towards this goal, Kurakin et al. (2017b) adversarially trained an Inception v3 model (Szegedy et al., 2016b) on ImageNet using a "single-step" attack based on a linearization of the model's loss (Goodfellow et al., 2014b). Their trained model is robust to single-step perturbations but remains vulnerable to more costly "multi-step" attacks. Yet, Kurakin et al. (2017b) found

*Part of the work was done while the author was at Google Brain.

that these attacks fail to reliably transfer between models, and thus concluded that the robustness of their model should extend to black-box adversaries. Surprisingly, we show that this is not the case.

We demonstrate, formally and empirically, that *adversarial training with single-step methods admits a degenerate global minimum*, wherein the model's loss can not be reliably approximated by a linear function. Specifically, we find that the model's decision surface exhibits sharp curvature near the data points, thus degrading attacks based on a single gradient computation. In addition to the model of Kurakin et al. (2017b), we reveal similar overfitting in an adversarially trained Inception ResNet v2 model (Szegedy et al., 2016a), and a variety of models trained on MNIST (LeCun et al., 1998).

We harness this result in two ways. First, we show that adversarially trained models using single-step methods remain vulnerable to simple attacks. For black-box adversaries, we find that perturbations crafted on an undefended model often transfer to an adversarially trained one. We also introduce a simple yet powerful single-step attack, which we call R+FGSM, that applies a small *random* perturbation—to escape the non-smooth vicinity of the data point—before linearizing the model's loss. While seemingly weaker than the Fast Gradient Sign Method of Goodfellow et al. (2014b), our attack significantly outperforms it for a same perturbation norm, for models trained *with or without adversarial training*.

Second, we propose *Ensemble Adversarial Training*, a training methodology that incorporates perturbed inputs transferred from other *pre-trained models*. Our approach *decouples* adversarial example generation from the parameters of the trained model, and increases the *diversity* of perturbations seen during training. We train Inception v3 and Inception ResNet v2 models on ImageNet that exhibit increased robustness to adversarial examples transferred from other holdout models, using various single-step and multi-step attacks (Goodfellow et al., 2014b; Carlini & Wagner, 2017a; Kurakin et al., 2017a; Madry et al., 2017). We also show that our methods globally reduce the *dimensionality* of the space of adversarial examples (Tramèr et al., 2017). Our Inception ResNet v2 model won the first round of the NIPS 2017 competition on Defenses Against Adversarial Attacks (Kurakin et al., 2017c), where it was evaluated on other competitors' attacks in a black-box setting.[1]

1.1 Subsequent Work (added April 2020)

Starting with the NIPS 2017 competition on Defenses Against Adversarial Attacks, many subsequent works have proposed more elaborate black-box transfer-based attacks. By incorporating additional optimization techniques (e.g., momentum (Dong et al., 2018), data-augmentation (Xie et al., 2019b; Dong et al., 2019) or gradients from a residual network's "skip connections" (Wu et al., 2020)), these attacks substantially improve the transferability of adversarial examples. While Ensemble Adversarial Training still improves a model's robustness to these attacks, the best attack to date—by Wu et al. (2020)—reduces the accuracy of our most robust ImageNet model to 22% (for ℓ_∞ perturbations bounded by $\epsilon = 16/255$).

We further note that adversarial training with multi-step attacks has now been scaled to ImageNet (Xie et al., 2019a), resulting in models with plausible and non-negligible robustness to *white-box attacks*, thereby superseding the results obtained with Ensemble Adversarial Training. Adversarial training with multi-step attacks is currently regarded as the state-of-the-art approach for attaining robustness to a fixed type of perturbations, whether in a white-box or black-box setting.

At the same time, a surprising recent result of Wong et al. (2020) suggests that with appropriate step-size tuning and early-stopping, adversarial training with the *single-step* R+FGSM attack yields models with *white-box* robustness that is comparable to that obtained with more expensive multi-step attacks (Madry et al., 2017).

2 Related Work

Various defensive techniques against adversarial examples in deep neural networks have been proposed (Gu & Rigazio, 2014; Luo et al., 2015; Papernot et al., 2016c; Nayebi & Ganguli, 2017; Cisse

[1]We publicly released our model after the first round, and it could thereafter be targeted using white-box attacks. Nevertheless, a majority of the top submissions in the final round, e.g. (Xie et al., 2018) built upon our released model.

et al., 2017) and many remain vulnerable to adaptive attackers (Carlini & Wagner, 2017a;b; Baluja & Fischer, 2017). Adversarial training (Szegedy et al., 2013; Goodfellow et al., 2014b; Kurakin et al., 2017b; Madry et al., 2017) appears to hold the greatest promise for learning robust models.

Madry et al. (2017) show that adversarial training on MNIST yields models that are robust to white-box attacks, if the adversarial examples used in training closely maximize the model's loss. Moreover, recent works by Sinha et al. (2018), Raghunathan et al. (2018) and Kolter & Wong (2017) even succeed in providing *certifiable* robustness for small perturbations on MNIST. As we argue in Appendix C, the MNIST dataset is peculiar in that there exists a simple "closed-form" *denoising* procedure (namely feature binarization) which leads to similarly robust models *without adversarial training*. This may explain why robustness to white-box attacks is hard to scale to tasks such as ImageNet (Kurakin et al., 2017b). We believe that the existence of a simple robust baseline for MNIST can be useful for understanding some *limitations* of adversarial training techniques.

Szegedy et al. (2013) found that adversarial examples transfer between models, thus enabling black-box attacks on deployed models. Papernot et al. (2017) showed that black-box attacks could succeed with no access to training data, by exploiting the target model's predictions to *extract* (Tramèr et al., 2016) a surrogate model. Some prior works have hinted that adversarially trained models may remain vulnerable to black-box attacks: Goodfellow et al. (2014b) found that an adversarial maxout network on MNIST has slightly higher error on transferred examples than on white-box examples. Papernot et al. (2017) further showed that a model trained on small perturbations can be evaded by transferring perturbations of larger magnitude. Our finding that adversarial training degrades the accuracy of linear approximations of the model's loss is as an instance of a *gradient-masking* phenomenon (Papernot et al., 2016b), which affects other defensive techniques (Papernot et al., 2016c; Carlini & Wagner, 2017a; Nayebi & Ganguli, 2017; Brendel & Bethge, 2017; Athalye et al., 2018).

3 The Adversarial Training Framework

We consider a classification task with data $x \in [0,1]^d$ and labels $y_{\text{true}} \in \mathbb{Z}_k$ sampled from a distribution $\mathcal{D}$. We identify a model with an hypothesis h from a space $\mathcal{H}$. On input x, the model outputs class scores $h(x) \in \mathbb{R}^k$. The loss function used to train the model, e.g., cross-entropy, is $L(h(x), y)$.

3.1 Threat Model

For some target model $h \in \mathcal{H}$ and inputs (x, y_{true}) the adversary's goal is to find an *adversarial example* x^{adv} such that x^{adv} and x are "close" yet the model misclassifies x^{adv}. We consider the well-studied class of ℓ_∞ bounded adversaries (Goodfellow et al., 2014b; Madry et al., 2017) that, given some *budget* ϵ, output examples x^{adv} where $\|x^{\text{adv}} - x\|_\infty \leq \epsilon$. As we comment in Appendix C.1, ℓ_∞ robustness is of course not an end-goal for secure ML. We use this standard model to showcase limitations of prior adversarial training methods, and evaluate our proposed improvements.

We distinguish between *white-box* adversaries that have access to the target model's parameters (i.e., h), and *black-box* adversaries with only partial information about the model's inner workings. Formal definitions for these adversaries are in Appendix A. Although security against white-box attacks is the stronger notion (and the one we ideally want ML models to achieve), black-box security is a reasonable and more tractable goal for deployed ML models.

3.2 Adversarial Training

Following Madry et al. (2017), we consider an adversarial variant of standard Empirical Risk Minimization (ERM), where our aim is to minimize the risk over adversarial examples:

$$h^* = \arg\min_{h \in \mathcal{H}} \quad \mathop{\mathbb{E}}_{(x, y_{\text{true}}) \sim \mathcal{D}} \left[\max_{\|x^{\text{adv}} - x\|_\infty \leq \epsilon} L(h(x^{\text{adv}}), y_{\text{true}}) \right] . \tag{1}$$

Madry et al. (2017) argue that adversarial training has a natural interpretation in this context, where a given attack (see below) is used to approximate solutions to the inner maximization problem, and the outer minimization problem corresponds to training over these examples. Note that the original